Versatile Neural Activation Predictor With Axon Structure Tailoring Capability Enabling Personalized Neuromodulation Computation

Hongda Li, Shunjing Wang, Xuesong Luo, Changqing Jiang, *Member, IEEE*, and Boyang Zhang

***Abstract*— Neuromodulation therapies are evolving to be more and more intelligent and personalized, driving the need for more precise and efficient stimulation strategies. Biophysically detailed computational models integrated with anatomically accurate neural structures could offer critical insights into neural activation patterns under various stimulation conditions, which are essential to optimize the treatment. However, solving these models containing a large number of nerve fibers is computationally intensive, especially when the neural targets comprise heterogenous axons, e.g., with varying geometries. Also, current methods lack generalizability across various neuromodulation scenarios, limiting the scalability and clinical utility of such models. In this study, we present a convolutional neural network (CNN)-based framework as a universal, rapid, and accurate alternative to conventional case-by-case brutal force computation methods. Our approach achieves a mean absolute error (MAE) of 6.91 $\times$ 10^{-3} mV and over 95% prediction accuracy under diverse extracellular stimulation scenarios, facilitating personalized simulations and tailored neuromodulation treatments.**

***Index Terms*— Neuromodulation, MRG model, activation thresholds, CNN.**

I. Introduction

NEUROMODULATION therapies have emerged as a critical approach in the treatment of neurological disorders [1], [2], [3], [4], [5], [6], [7], [8]. By delivering electrical stimulation via percutaneous or implanted electrodes to increase neuronal excitability in target areas, these therapies can modulate neural activity and alleviate symptoms. With the advancement of therapeutic technologies, neuromodulation is evolving toward increased intelligence and personalization. For instance, personalized epidural electrical stimulation (EES) strategies have shown promise in restoring lower-limb motor function in patients with spinal cord injury [5], [6], [7], [9]. However, substantial inter-individual variability presents a major challenge, as achieving optimal therapeutic outcomes often requires fine-tuned selection of stimulation parameters [9], [10]. Similar issues also arise in other neuromodulation therapies, such as deep brain stimulation (DBS) [2] and vagus nerve stimulation (VNS) [11], where treatment efficacy is closely tied to parameter optimization tailored to individual patient characteristics.

Computational modeling has been proved to be a powerful tool for investigating the mechanisms of neuromodulation and optimizing stimulation strategies [12], [13], [14], [15], [16], [17]. By precisely modeling the geometry of neural structures and interfaces and performing simulations, researchers can derive the electric field distributions generated by neuromodulation using the finite element method (FEM) [6], [15]. These calculated electric field distributions enable the estimation of nerve fiber responses, which could guide the design of optimized neural interfaces [9], the selection of stimulation parameters [18], and the development of personalized or closed-loop treatment plans [7], [9], [19].

The multi-compartment double-cable model is currently one of the most accurate methods for simulating neural activities of myelinated fibers [20]. By modeling the intricate structure of nerve fibers and the properties of ion channels, accurate simulations of membrane potential changes in response to extracellular electrical stimulation can be achieved [20], [21], [22]. However, calculating the dynamic responses of nerve fibers in complex electric fields requires considerable computational resources, limiting the practical application of these models. Simplified approaches, such as the calculation of the volume of tissue activated (VTA), have been widely adopted in the DBS field to estimate the impact range of electrical stimulation [23], [24], [25]. While these methods provide useful guidance for determining stimulation locations and parameters, they usually overlook the specific trajectories of nerve fibers, resulting in deviations from actual physiological conditions [26], [27]. Another simplified method is using driving-force (DF) models to replace the computation of multi-compartment cable models. The activating function

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(i.e., second-order spatial derivative of the electric potential along the axon) is used to predict activation thresholds of the nerve fibers [28]. More advanced studies utilize the weighted sum of activating functions at the nodes of the axon to achieve higher accuracies [29], [30], [31]. However, this simplification might struggle to accurately depict the complex nonlinear behaviors inherent in the cable models, leading to poor performance under certain stimulation scenarios [31]. A recent work reduced the computational burden by simplifying the axon's physical model. However, it still involved solving a large number of partial differential equations [32].

Recent advances in machine learning offer an alternative solution. Artificial neural networks (ANNs) have been used to predict the VTA from axon diameter and stimulation configurations [33]. Other studies have also demonstrated that ANNs [34] and convolutional neural networks (CNNs) [35], [36] can predict the activation thresholds of nerve fibers under varying electric field distributions, potentially replacing time-consuming NEURON simulations [37]. However, existing models face challenges in handling nerve fibers of different lengths or diameters, and their application scenarios remain constrained due to the limitation of training data. With the advances in neuromodulation strategies and neural interfaces, the potential distribution along the axon could be irregular because of more complex stimulation configurations. It has been reported that nodes located more than five internodes away from the node of interest contribute negligibly to the axonal response under extracellular stimulation [29]. Based on this, some studies estimate the activation threshold by evaluating 11 nodes surrounding the region with the minimum extracellular potential, and use this value as a proxy for the activation threshold of the entire axon [34]. However, this method is inadequate in scenarios where action potentials initiate at one end of the axon [38].

Further, the axon model's geometric parameters were originally published for a discrete list of fiber diameters [20], [21], [22], which leads to inconvenience when modeling the axons with diameters out of the list. Some studies have developed new model files by fitting the geometric parameters of various axonal microstructures (e.g., internode length, node diameter). However, this method can lead to activation threshold estimation errors exceeding 10% [39].

In response to the growing demand for more precise neuromodulation, it is essential to develop a more flexible and efficient computational approach for modeling neural responses, which can facilitate fine-tuning of stimulation parameters and enhance rehabilitation outcomes. In this study, we propose a universal CNN-based computational framework capable of predicting activation thresholds for myelinated axons. By training and testing the neural network with a dataset combining randomly generated and FEM calculated potential distributions, we established the capability to accurately estimate activation states of fibers under various stimulation configurations, such as complex contact settings and varying pulse widths. Furthermore, the framework's applicability extends to diverse fiber geometries (lengths and diameters), significantly enhancing its versatility. We validated the framework's predictive capability with a personalized EES model, achieving 96.9% accuracy while significantly reducing the time cost by three to four orders of magnitude compared to traditional methods. The similar performance in DBS scenario demonstrates the generalizability of the proposed model to other neuromodulation therapies. Besides, using the axon diameter as an independent input dimension, the proposed computational framework enables the inclusion of a broader range of axon diameters without the necessity of generating new model files or performing additional simulations. This approach offers an efficient and robust alternative to conventional simulations, facilitating rapid parameter selection and supporting the development of personalized and intelligent neuromodulation strategies.

II. Methods

A. Combined Dataset for Machine Learning

To develop a robust capability for accurately predicting nerve fiber activation thresholds, we first constructed a combined dataset consisting of 400,000 entries to train the neural network (Fig. 1a, b). Each entry in the dataset contained key variables: the potential distribution along the axon, fiber length, fiber diameter, stimulation pulse width, and the corresponding activation threshold calculated with NEURON [37] (Fig. 1d). To comprehensively simulate the potential fields along axon trajectories in different neuromodulation scenarios, we employed two distinct methods for generating the potential distributions:

Random Generation: 200,000 potential distributions were randomly generated. First, 3 to 6 points were randomly selected within a 1×1 space, whose horizontal coordinate represents the position along the axon trajectory, and vertical coordinate represents the electric potential. Then we used a fifth-order spline to interpolate a continuous potential-position curve from the selected points. Finally, both the horizontal and vertical coordinates were normalized to ensure consistency in the whole dataset.

Finite Element Method (FEM) Calculation: An additional 200,000 potential distributions were obtained using the finite element method (FEM), as described in detail in the "Model and Simulation of EES" section. This method provided physical realistic potential distributions generated by complex stimulation configurations in neuromodulation scenarios. The calculated potential distributions were also normalized.

For each potential distribution, we randomly assigned a fiber length (ranging from 20 to 100 mm), diameter (chosen from 5.7, 7.3, 8.7, 10.0, 11.5, 12.8, 14.0, 15.0, or 16.0 μm), and stimulation pulse width (ranging between 100 and 1000 μs), which significantly affect activation thresholds of nerve fibers (Fig. 1c). The selection range of these parameters could cover most neuromodulation scenarios.

We then implemented a multi-compartment double-cable model based on the MRG (McIntyre-Richardson-Grill) model to simulate the nerve fiber's response [21], which is the current gold standard for modeling myelinated fibers [40], [41]. This model features a double-cable conductance-based compartmental design, rooted in Blight's foundational architecture

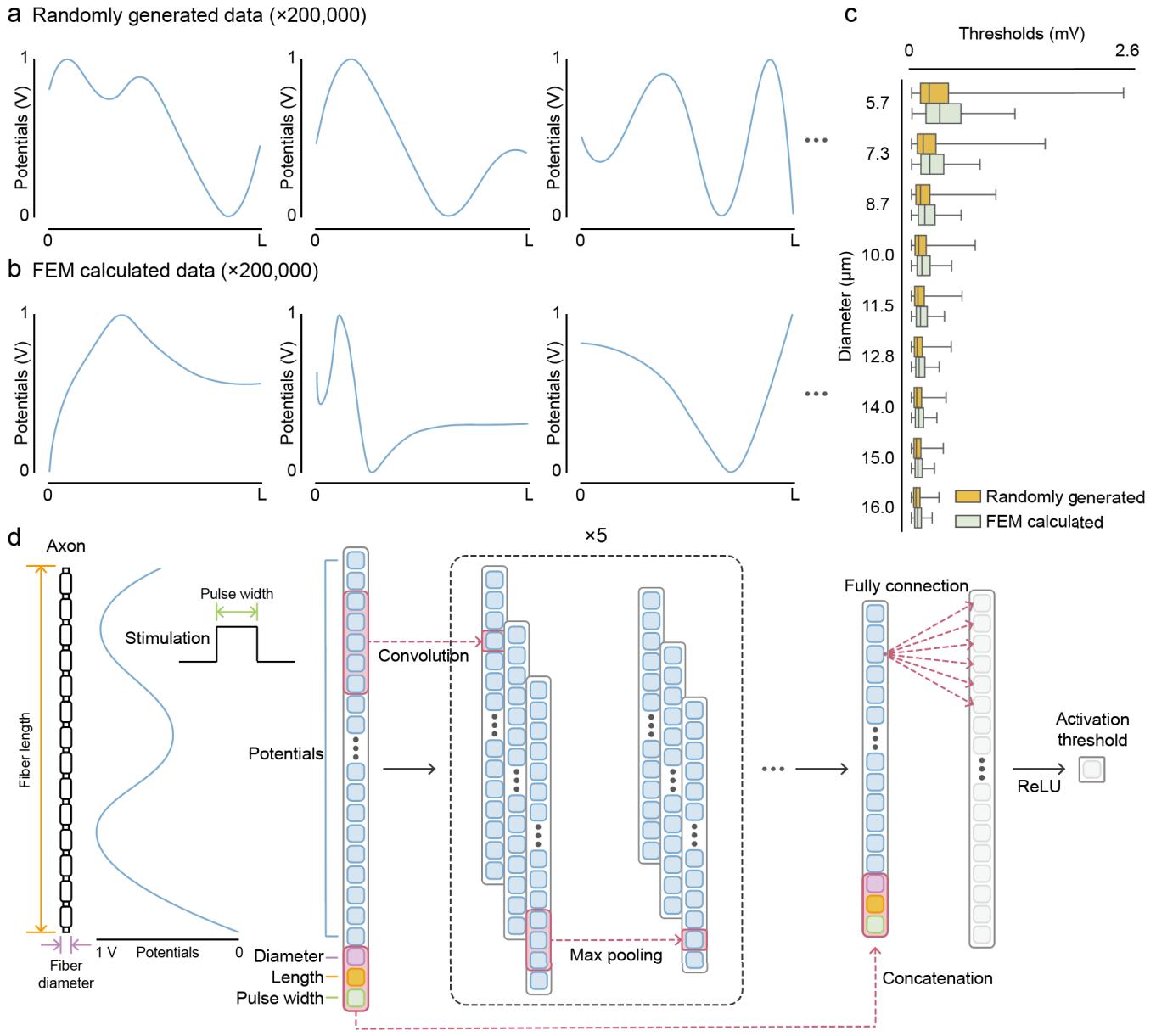


Fig. 1. Composition of the dataset and structure of the neural network. The dataset used for training and testing the neural network consists of 200,000 randomly generated items and 200,000 items calculated from FEM calculation. (a), Three potential distributions randomly generated. (b), Three potential distributions calculated from EES simulation. (c), Distribution of activation thresholds of the dataset used. For each diameter, the top 10% with the highest thresholds are not shown in this figure. (d), Structure of the neural network. The network consists of five groups of convolution-max pooling layers and two fully connected layers. Fiber diameters, fiber lengths and pulse widths are concatenated with the convolved potential distributions before fully connected layers. Activation thresholds are predicted using ReLU as the activation function.

and enhanced by the representation of paranodal ultrastructure [42], [43]. Non-nodal regions are treated as passive, while nodal regions incorporate active Hodgkin-Huxley-like conductance [44]. The outer cable, which simulates the myelin sheath, is comprised entirely of passive compartments [21]. We obtained the model files from ModelDB with accession number 3810 [45]. Using NEURON simulation environment [37], we computed the electrical response of MRG models under single-pulse extracellular stimulation for all the generated entries. We interpolated the potentials for each node of the axon from the potential-position curves and used the "play" method to apply the stimulation sequence to the axon. The simulation was conducted with a time step of 5 μs. The criterion for activation was defined as the successful propagation of an action potential along the entire length of the axon [16]. We determined the activation threshold using a binary search algorithm, refining the threshold with an error tolerance of less than 1×10^{-7} V. The extracellular potential distribution along each axon was normalized by setting the maximum potential to 1 mV and the minimum to 0 mV. When computing the axon activation threshold in NEURON, the normalized potential profile is multiplied by a scaling factor and applied to the axon. The scaling factor at which the axon is just activated is considered the activation threshold,

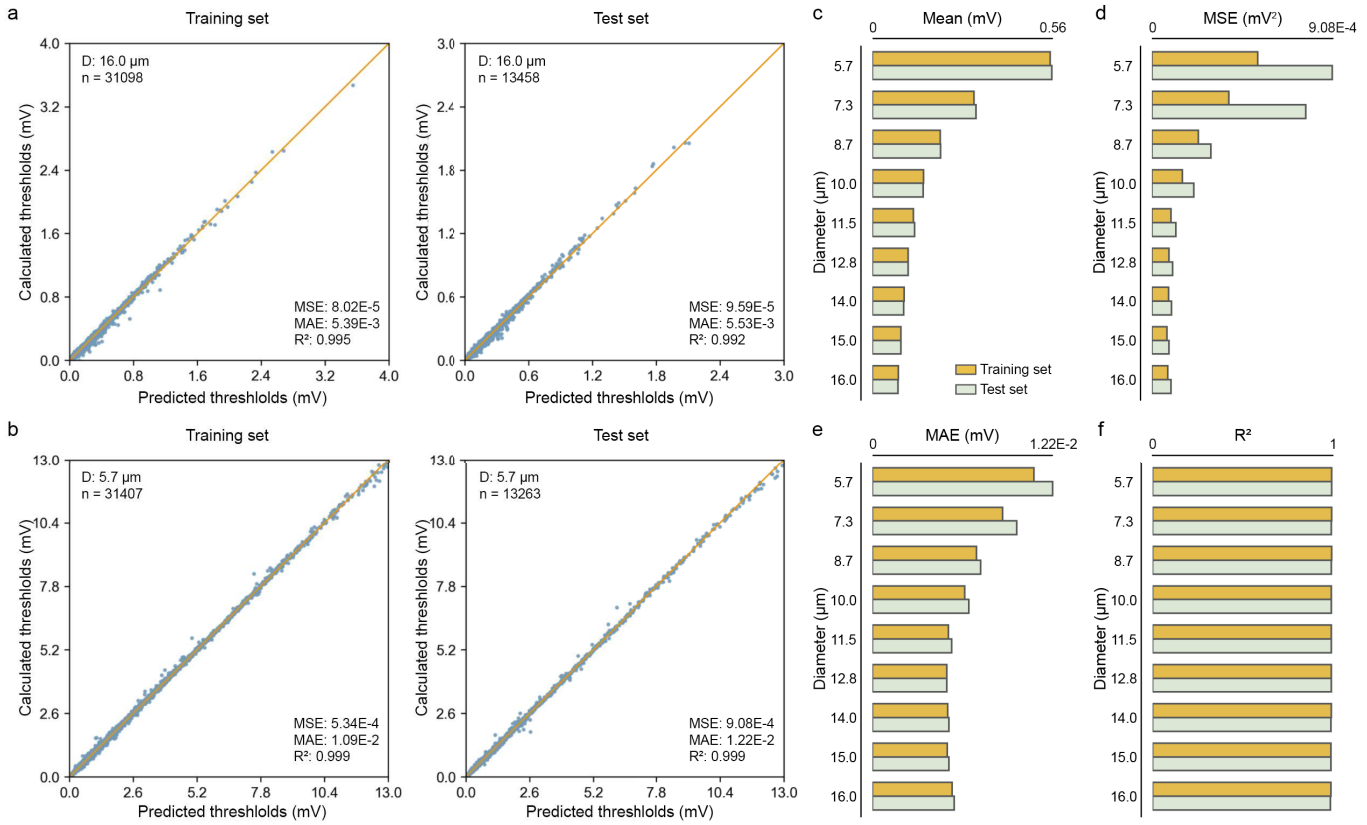


Fig. 2. Performance of the CNN-based computation framework. (a), Threshold prediction performance for nerve fibers with a diameter of 16.0 μm. (b), Threshold prediction performance for nerve fibers with a diameter of 5.7 μm. (c), Mean activation thresholds of nerve fibers with different diameters. (d), Mean Squared Error of activation thresholds for nerve fibers with different diameters. (e), Mean Absolute Error of activation thresholds for nerve fibers with different diameters. (f), R2 of activation thresholds for nerve fibers with different diameters.

representing the minimum electric field strength required to elicit an action potential in the axon. The simulation results demonstrated that nerve fibers with larger diameters have relatively lower activation thresholds (Fig. 1c).

B. CNN-Based Predictor

In this study, we utilized TensorFlow [46] (version 2.17) in Python to construct, train, and test a convolutional neural network (CNN) [47] (Fig. 1d). The input to the network consisted of a 1×100 vector representing the normalized potential distribution along the axon trajectory, and three other normalized dimensions representing the diameter, length of the nerve fibers, and the stimulation pulse width. A total of five convolutional-pooling layers were employed to efficiently extract and integrate features from the potential distribution, which were then concatenated with the other three other input features. Two fully-connected layers were subsequently used to predict the activation threshold, with rectified linear units (ReLU) applied as the activation function.

The dataset, comprising randomly generated entries and FEM calculations, was randomly divided into a training set (70%) and a test set (30%). The neural network was trained using the default fit framework in TensorFlow. To prevent overfitting, ten-fold cross-validation was implemented, whereby the training set was partitioned into ten subsets, with one subset serving as the validation set in each iteration. The Mean Squared Error (MSE) on the validation set was employed as the loss function, and training was terminated if there was no improvement in validation performance over 30 epochs. The Adam optimizer was utilized with a learning rate of 0.0003. Ultimately, the model exhibiting the best performance across the ten validation sets was saved. To derive the final predicted activation threshold, an ensemble method was adopted, whereby the average of the predictions from the ten models was used as the final output [48]. This approach effectively mitigates validation set bias and enhances the robustness of the predictions. The trained models were subsequently validated on the test set using metrics including Mean Absolute Error (MAE), MSE, and R^2 (Fig. 2). Notably, the test set was completely disjoint from both the training and validation sets.

C. Validation of the Model's Predictive Capability in Diameter Dimension

To verify the capability of the model to predict activation thresholds of axons with diameters not contained in the training set, we retrained a CNN with the same network structure using the entries with 5 specific diameters (5.7, 8.7, 11.5, 14.0, 16.0 μm) from the 400,000 entries. We divided the new dataset (containing 222,410 entries) into a training set (70%) and a test set (30%) and employed the same training techniques to obtain

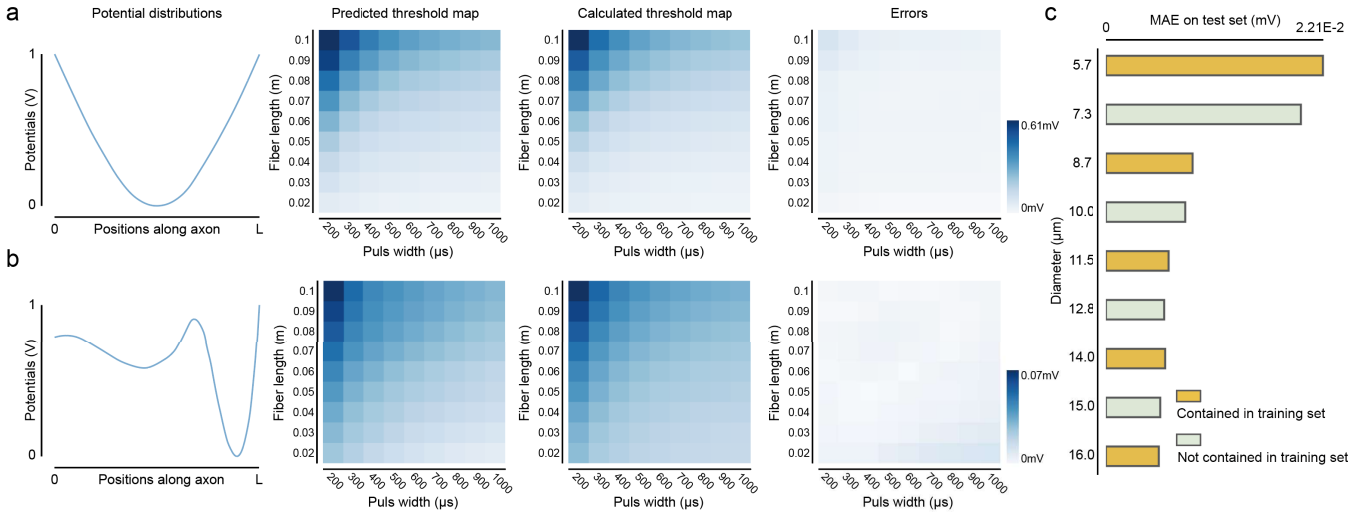


Fig. 3. Validation of model's predictive capability in multiple dimensions. (a), Threshold maps of a FEM calculated potential distribution. The potential distribution was obtained from FEM calculation under a typical unipolar stimulation. Left: the threshold map predicted by CNN. Middle: the threshold map calculated by NEURON. Right, the error map between prediction and NEURON calculation. (b), Threshold maps of a randomly generated potential distribution. (c), Validation of model's predictive capability using datasets with axon diameters not contained in the training set.

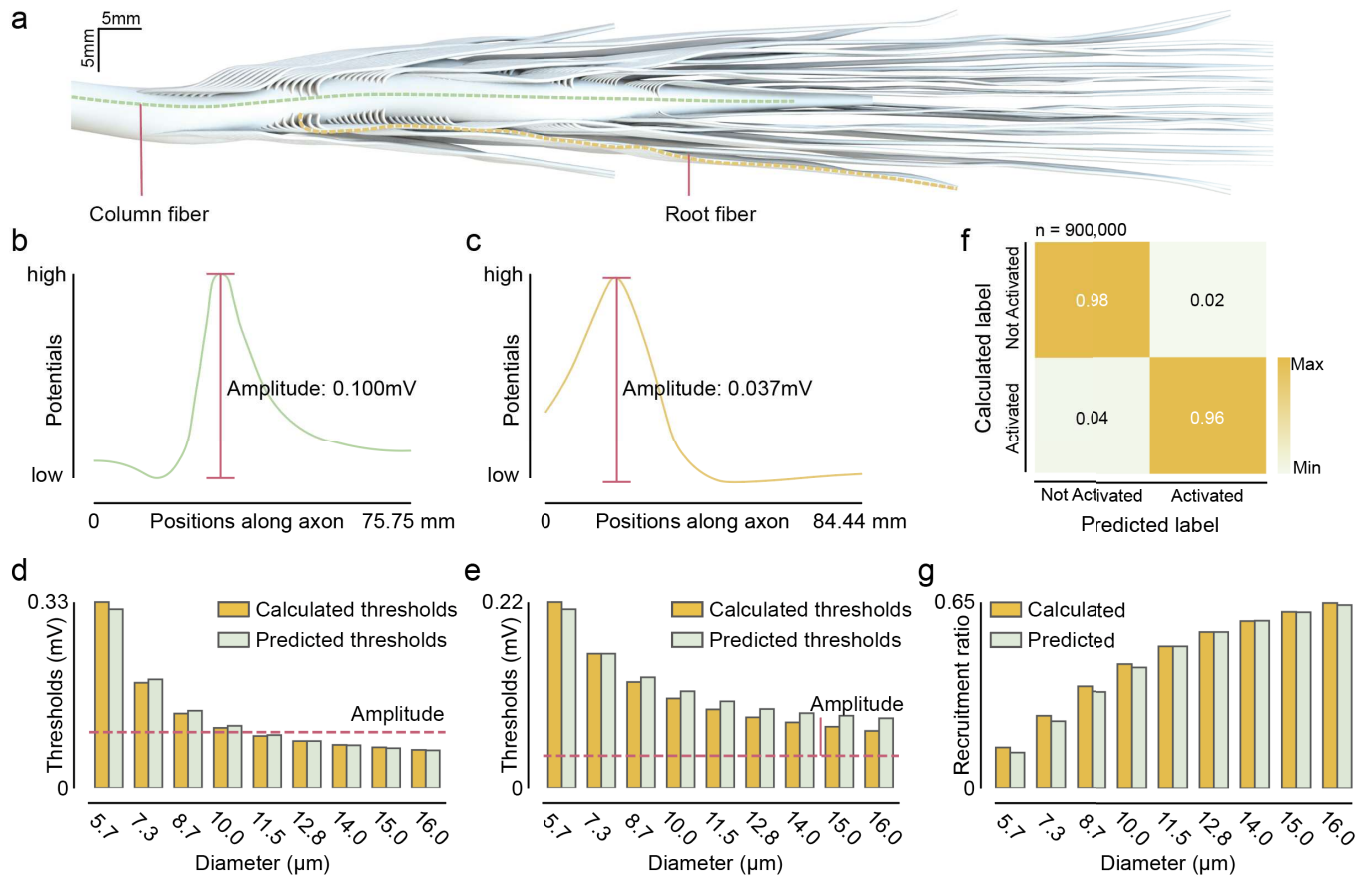


Fig. 4. Validation of the CNN-based estimator under EES scenario. (a), Illustration of the spinal cord model. We generated two kinds of fiber trajectories: column fibers (170 trajectories) in white matter and root fibers (320 trajectories) in spinal nerve roots. (b), Example of potential distributions of a column fiber. (c), Example of potential distributions of a root fiber. (d), Predicted thresholds and calculated thresholds of the column fiber in (b). (e), Predicted thresholds and calculated thresholds of the root fiber in (c). (f) Confusion matrix of predicted activation states and calculated activation states for 900,000 fibers under EES simulation. (g) Calculated and predicted recruitment ratios for 900,000 fibers in EES scenario.

a new CNN. We used the entries with the other 4 diameters (7.3, 10.0, 12.8, 15.0 μm) as the validation set to check the interpolation capability of the CNN model in the diameter dimension. The validation set contains 177,590 entries, which are not included in the training process. We employed the CNN to predict the activation thresholds and compared with the calculated thresholds using NEURON.

D. Personalized Modeling and Simulation of EES

We developed a personalized spinal cord model to obtain the potential distribution of nerve fibers under actual electrical stimulation and to validate the model's predictive accuracy. An anatomically precise geometric model of the spinal cord was constructed based on MRI data from a participant [8], [49], encompassing the gray matter, white matter, cerebrospinal fluid, dura mater, and nerve roots for the T12-S2 segments (Fig. 4a). The contours of the spinal cord's white matter and cerebrospinal fluid were inferred from axial MRI images, while the gray matter contours were derived from established anatomical atlases [50]. Fiber trajectories for the dorsal and ventral roots were traced and interpolated from MRI images at various locations. For each nerve root, 10 fiber trajectories were generated to simulate the "bundle" structure at the junction between the nerve roots and the spinal cord, with starting points uniformly distributed across the entry region of the spinal cord segment [9] (Fig. 4a).

To achieve a uniform distribution of fibers throughout the white matter, we modified the Lloyd relaxation method [51]. Several cross-sections of the white matter were selected uniformly along the axial direction. For the most rostral section, we applied Lloyd relaxation to uniformly redistribute randomly scattered points within the white matter area. These redistributed points were then re-centralized and used as initial seeds for the Lloyd relaxation algorithm in subsequent cross-sections. Any initial seeds located outside the white matter in new sections were removed, and the remaining points were redistributed across all sections. Points sharing the same index were interpolated to generate nerve fiber trajectories. The number of scattered points in the most rostral section was adjusted to achieve the desired number of axons In this model, we obtained 170 trajectories of nerve fibers within the white matter. The model was constructed and assembled using SolidWorks software, and a 32-contact spinal epidural electrode was designed and positioned in the epidural space, targeting the lumbar enlargement. The whole model was placed in a saline cylinder to mimic the in vivo environment [6], [16]. The completed model was imported into COMSOL Multiphysics for mesh generation and finite element analysis. Conductivity values were obtained from established data and assigned to different tissues [52] (Table S1).

The anodes and cathodes for the EES were selected randomly within the 32 contacts. The number of cathodes varied between 1 and 3, while the number of anodes ranged from 1 to 5. Stimulation amplitudes were selected between 3 and 5 mA with pulse widths ranging from 100 to 1000 microseconds, resulting in a total of 1800 stimulation configurations. For each configuration, the corresponding electric field was computed using COMSOL Multiphysics, and the potential distributions along 490 nerve fiber trajectories (170 trajectories in white matter, 320 trajectories in nerve roots) were recorded. To ensure a roughly balanced number of activated and non-activated nerve fibers, all computed potential values were amplified by a factor of 20, which is reasonable under quasi-static assumptions [53]. This process yielded over 800,000 potential distributions, from which 200,000 were randomly selected as part of the dataset for training and testing the neural network. Another 10,000 potential distributions were used to form an additional test set. For each potential distribution, we calculated the neural responses of MGR models with 9 different diameters, resulting in a dataset containing 900,000 entries. Examples of calculated position-potential curves for trajectories in white matter and nerve roots are demonstrated in Fig. 4b and Fig. 4c, respectively. The activation states of the additional test sets were calculated based on actual lengths of fiber trajectories and stimulation pulse widths. In parallel, the electric potential distributions along the fiber trajectories were normalized and utilized, alongside the fiber diameter, length, and stimulus pulse width, as inputs to the trained neural network for predicting the corresponding activation threshold. The predicted activation thresholds were then compared to the actual stimulus amplitudes. If the activation threshold was found to be lower than the scaling factor applied during normalization, the nerve fiber was predicted to be activated; otherwise, it was deemed not activated (Fig. 4d, e). Confusion matrices of predicted and calculated activation states were calculated for fibers with different diameters (Fig. 4f). Predicted and calculated recruitment ratios were compared for axons with different diameters (Fig. 4g).

E. Modeling and Simulation of DBS

We developed a simplified DBS model to further validate the accuracy of the model's prediction (Fig. 5a). Initially, we modeled a DBS directional electrode comprising eight contacts. Personalized neural fiber trajectories [54], derived from diffusion tensor imaging (DTI), were integrated with the DBS electrode within a cubic bulk environment (side length 0.1 m, conductivity 0.2 S/m) [31], [34]. Additionally, a layer of encapsulation tissue (0.5 mm thick, conductivity 0.07 S/m) was modeled around the electrode [55]. The assembled model was import to COMSOL Multiphysics and a nonuniform, second-order tetrahedral mesh was generated for the discretization.

We randomly generated 10 distinct stimulation configurations, encompassing different cathodes, anodes, stimulation intensity, and pulse width, and utilized COMSOL Multiphysics to calculate the corresponding electric fields (Fig. 5b). The anodes were set to be current sources and the cathodes were set to be grounded. Following the FEM calculation, the activation states of acoustic radiation pathway (AR, containing 2920 trajectories) and corticostriatal pathway (CS, containing 2252 trajectories) fibers were calculated using the NEURON simulation environment [54]. The calculated potentials were amplified with a factor of 7 to make sure the total recruitment ratio was around 0.5. In a manner similar to EES, we

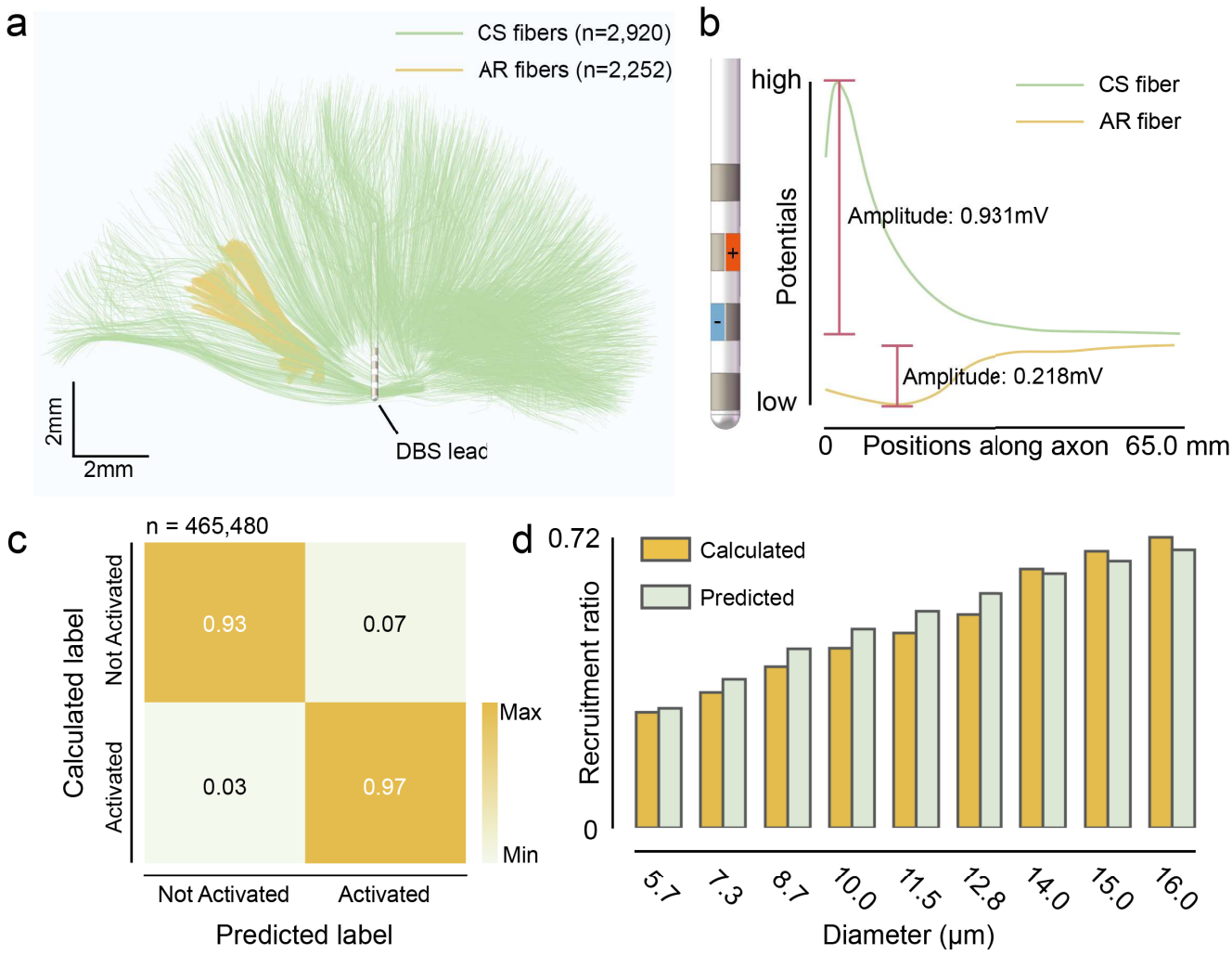


Fig. 5. Validation of the CNN-based estimator under DBS scenario. (a), Illustration of the DBS model. We selected two groups of brain nerve fibers as the subjects of the computational study. The DBS lead was placed among these fibers. (b), Examples of potentials distributions of a CS fiber and a AR fiber. The DBS configuration was shown in the left panel. (c), Confusion matrices of predicted activation states and calculated activation states for 465,480 fibers. Ten DBS configuration were generated randomly. For each fiber trajectory, the activation states of nerve fibers with 9 different diameters were predicted and calculated separately. (d), Predicted and calculated Recruitment ratios of nerve fibers of different diameters.

employed the trained neural network to predict the activation thresholds for each neural fiber and assess their activation states. Activation thresholds were predicted by the CNN model and predicted activation states were judged according to whether stimulation amplitudes were greater than the predicted activation thresholds. Confusion matrices of predicted and calculated activation states and recruitment ratios were calculated for fibers in different groups (Fig. 5c, d).

F. Time Cost Comparison

To quantitatively assess and compare the computational efficiency of CNN-based activation state prediction versus NEURON simulations, we focused our analysis on neural fibers of 10 centimeters in length (Fig. 6a). For fibers with different diameters, we simulated neural responses under 100 distinct, randomly generated stimulation configurations. These simulations were executed within the NEURON simulation environment, with a simulation duration of 7 ms (sufficient to capture the complete propagation of the action potential), a temporal resolution of 5 μs, and utilizing a single CPU for processing. The total time required for each simulation was meticulously recorded.

In parallel, we employed the same computational platform to perform CNN-based threshold predictions and activation assessments. We adjusted the input dimensions to comprehensively assess the performance of the CNN model across varying scales.

To evaluate the time efficiency and consistency in practical applications of both methods, we systematically recorded the total time required to determine the activation states of 1960 neural fibers (containing nerve fibers with diameters of 16.0/10.0/7.3/5.7 μm representing Aα/Aβ/Aγ/Aδ fiber groups) for a single, randomly chosen EES parameter [56] (Fig. 6b). This process was replicated ten times, each time with a different EES parameter, to enable the calculation of the mean

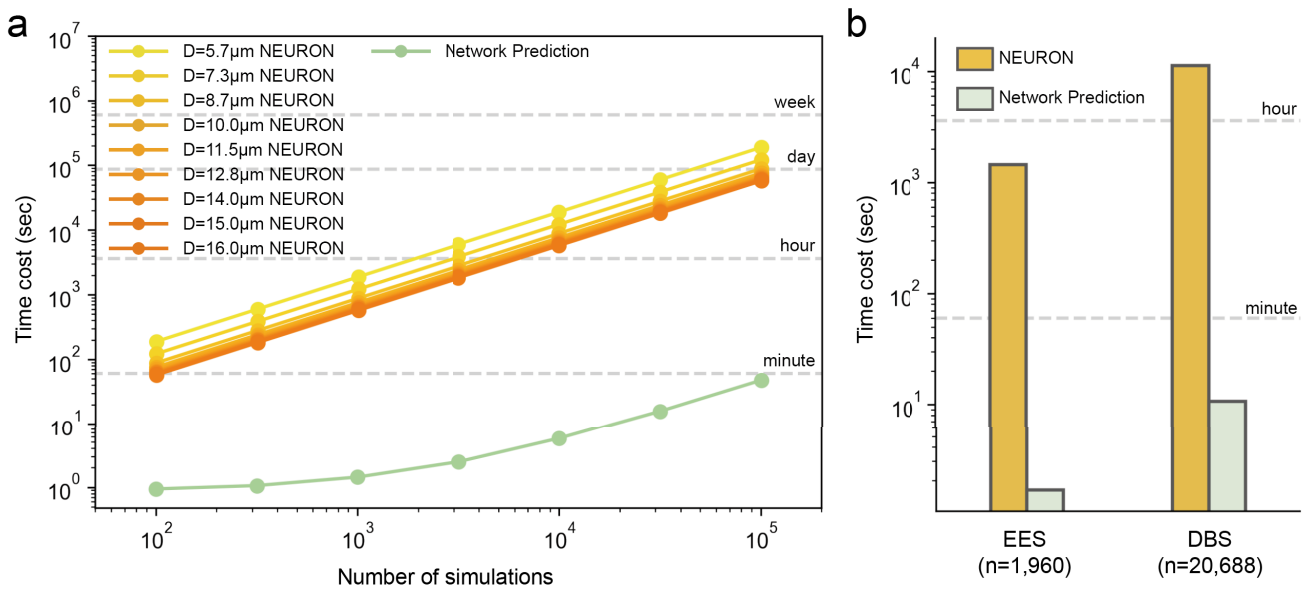


Fig. 6. Time cost of activation states calculation using CNN-based framework and NEURON. (a), The time spent calculating activation states in different methods using the same device. Simulation of nerve fibers with smaller diameters requires more time. (b), Time cost comparison for calculating activation states under a single configuration in EES and DBS scenarios.

and variance of the time consumption. Similarly, we applied the CNN-based computational framework to the same set of EES parameters and neural fibers, meticulously recording the corresponding time expenditures. We also compared the time cost to predict the activation states of 20,688 fibers in DBS simulation for each stimulation configuration (Fig. 6b).

III. Results

A. Accurate and Robust Prediction of Activation Thresholds for MRG Models Under Different Stimulation

Through rigorous training on the comprehensive dataset that integrates both randomly generated and FEM-calculated potential distribution data (Fig. 1), our CNN-based prediction framework demonstrated remarkable accuracy in predicting activation thresholds for neural fibers of diverse lengths and diameters under varying stimulation conditions (Fig. 2). The model's performance remained consistent across both the training and testing sets, showcasing its robustness and reliability.

Specifically, the overall MAE on the testing set was an impressive 6.91×10^{-3} mV, with a mean squared error (MSE) of 2.96×10^{-4} mV, indicating the high-performance predictive capability. It is noteworthy that, due to the inherent nature of neural fibers where smaller diameters correlate with higher activation thresholds, the prediction errors for small-diameter fibers (e.g., 5.7 μm) were slightly higher (mean of 0.55 mV, MAE of 1.22×10^{-2} mV) compared to those for larger-diameter fibers (e.g., 16.0 μm, with a mean of 0.079 mV, an MAE of 5.53×10^{-3} mV) (Fig. 2). Non-theless, the strong linear correlation between the predicted and actual activation thresholds, as evidenced by R2 values consistently exceeding 0.99 (Fig. 2f). This result underscores the potential of our approach for accurate and efficient activation threshold predictions in neural fibers, with implications for advancing neural engineering and neuromodulation technologies. The related data and code have been uploaded to Google Drive. Further data are available upon reasonable request and can be obtained by contacting the corresponding author via email.

In order to assess the model's ability to accurately capture the influence of axon length and stimulation pulse width on activation thresholds, we constructed activation threshold maps by varying stimulation pulse widths and fiber lengths with fixed potential distributions (Fig. 3). We plotted activation threshold maps corresponding to a FEM calculated potential distribution (Fig. 3a) and a randomly generated potential distribution (Fig. 3b), neither of which was included in the training set. By assessing the differences between the predicted thresholds by CNN and computed thresholds by NEURON, we demonstrate that the CNN-based predictive framework effectively captures the influence on the activation thresholds in these two dimensions.

B. Expansion in Selection Range of Axon Diameter

We used axon diameters as an independent continuous input dimension to the neural network, which means activation thresholds of axons with various diameters out of the input list could be calculated. The model could accurately predict activation thresholds of axons with diameters not contained in the training set (7.3, 10.0, 12.8, 15.0 μm) with an average MAE of 9.84×10^{-3} mV, which is similarity to the MAE of 9.74×10^{-3} mV on the test set containing data of axons with 5 diameters (Fig. 3c). The results confirm the reliability of the CNN model in predicting activation thresholds for axons with new diameters, thereby expanding the applicable range of axon diameters in neural computational studies. The CNN model trained on a dataset comprising 9 diameters is capable of

predicting activation thresholds for nerve fibers with diameters beyond those included in the training set, without the need to reconstruct new fiber models.

C. Predicting Neural Recruitment in EES Scenario

Utilizing an extensive dataset of 100,000 EES simulation-derived potential distributions, which were entirely independent from the original training and testing sets, our CNN-based prediction framework was successfully applied to estimate the activation thresholds and accurately determine the activation states of neural fibers based solely on their individual lengths and stimulation pulse widths (Fig. 4).

Remarkably, the model demonstrated exceptional predictive accuracy across a wide range of fiber diameters, achieving an overall prediction accuracy exceeding 96.9% (Fig. 4f, g). Even in the rare instances where the predicted activation states deviated from the actual state, the estimated activation thresholds remained in close proximity to the true values, highlighting the model's precision and reliability. This outcome shows the practical applicability and efficiency of our CNN-based model, significantly reducing the computational burden compared with traditional methods.

D. Predicting Neural Recruitment in DBS Scenario

Despite the unique challenges posed by DBS electrodes, including their smaller inter-contact spacing and varying relative positions to neural fibers, our CNN-based computational framework achieved remarkable success in predicting and classifying the activation patterns of brain fibers under DBS (Fig. 5). Notably, these electric potential distributions were entirely novel and had not been included in any prior training or testing datasets.

The overall prediction accuracy surpassed 95.3%, with particularly high accuracies of 97.7% for AR (acoustic radiation) fibers and 93.5% for CS (corticostriatal pathway) fibers (Fig. 5c, d). Validation in new neuromodulation scenario proves the robustness and generalizability of the computational framework.

E. Significant Reduce of Time Cost

The CNN-based predictive framework dramatically reduces the time required to ascertain the activation states of nerve fibers under neuromodulation. Specifically, using a single CPU to calculate the response of 100 nerve fibers, each 10 cm long and 16 μm in diameter, takes approximately 60 seconds. However, for fibers of 5.7 μm in diameter, the computation time increases to around 200 seconds due to the higher number of nodes. In contrast, predictions made by the neural network take just 1 second. For a substantial dataset of 100,000 such fine fibers, the traditional single-CPU approach would exceed two days, whereas the CNN framework completes the task in under a minute (Fig. 6a).

Furthermore, when applied to the EES model encompassing 1960 axons, determining the nerve fiber recruitment ratio for each parameter via conventional methods consumes approximately 25 minutes. In comparison, neural network predictions are achieved in a mere 1.6 seconds, highlighting the substantial efficiency gains offered by our CNN-based approach (Fig. 6b). In DBS scenes containing more nerve fibers, it takes more than 3 hours to calculate their activation states under a single stimulation configuration using NEURON simulation, whereas it takes only 10 seconds using the CNN-based framework (Fig. 6b).

IV. Discussion

A. A Generalizable CNN Framework for Predicting Neural Activation

The MRG nerve fiber model is the most widely employed model for simulating neural responses, particularly in myelinated axons of mammals under extracellular electrical stimulation [40]. To enhance the efficiency of estimating the MRG model's responses across various scenarios, we developed a CNN-based framework that accurately predicts activation thresholds (Fig. 1). This approach significantly reduces computational time compared to traditional NEURON simulations while maintaining high prediction accuracy.

Unlike the VTA and DF methods, our framework considers the specificity of nerve fiber trajectories and geometries [26], which facilitates more detailed and more personalized modeling and simulation. Additionally, our method outperforms previous neural network-based approaches to threshold prediction by accommodating nerve fibers of varying lengths and diameters under arbitrary electrical stimulation [34]. This advancement enables the construction of a universal threshold prediction model suitable for diverse neuromodulation scenarios (Fig. 2). The accuracy exceeding 95% of activation prediction in EES and DBS scenarios has demonstrated the versatility of this computational framework, which means that the activation thresholds of nerve fibers with various diameters can be immediately estimated, given the potential distribution along the axonal trajectory and stimulation settings. This threshold estimator can be extended to many other neuromodulation scenarios involving interactions between axons and extracellular electrical stimulation, such as peripheral nerve electrical stimulation and transcranial direct current stimulation (tDCS). This method significantly reduces the time required to compute activation states while ensuring accuracy, providing favorable conditions for the systematic computation of models involving large population of nerve fibers.

B. Dataset and Network Features

Besides potential distributions calculated with FEM, we specially add the randomly generated parts to the training dataset, which allows the CNN to more effectively capture features from potential-position curves. This is the key operation to increase the model's applicability under complex stimulation scenarios. After convolution and pooling, we incorporated fiber diameter, length, and stimulation pulse width as input features of the fully connected layers (Fig. 1d), enhancing the neural network's ability to generate comprehensive threshold estimations.

We included axon length as one of the input variables to the CNN. Previous studies have shown that under simple stimulation conditions, action potential initiation is primarily determined by a few nodes near the stimulation site, while nodes far from the stimulation site contribute negligibly [29], [34]. In some stimulation scenarios, calculating the activation threshold based on 11 nodes surrounding either the node with the minimum extracellular potential or the node with the minimum second derivative of the potential has been shown to be a reliable approximation of the threshold for the entire axon [34]. However, the reliability of this method under more complex stimulation configurations remains to be further validated. In particular, it is not applicable when the action potential is initiated at one end of the axon.

C. Generalization Across Axon Diameters

In the MRG model, the selection of fiber diameters is limited [20], [22]. Different fiber diameters result in variations in the geometric dimensions of various model structures. For a given fiber trajectory, computing the response of axons with different diameters requires constructing multiple models and running multiple simulations. In the training process, we used discrete fiber diameters as inputs, but after training, the CNN was able to provide a more continuous range of options. In this study, fiber diameter was treated as a continuous input dimension, and the CNN effectively mapped it to activation thresholds. We conducted an experiment to verify the predictive capability for axons with new diameters (Fig. 3c). The accurate prediction of activation thresholds for axons with diameters not included in the training process implies the potential to give reliable predictions for axons with arbitrary diameters between 5.7 and 16.0 μm. We believe this advancement introduces a more valuable computational tool for neuromodulation research.

The diameters of nerve fibers can range from 0.2 μm to 20 μm [57]. To further assess the extrapolation capability of our model for predicting activation thresholds of fibers beyond the 5.7–16.0 μm diameter range, we conducted an additional computational experiment: we trained the CNN using only the intermediate seven diameters, while keeping the smallest (5.7 μm) and largest (16.0 μm) diameters as the test set (Fig. S2). The results showed that the trained model could still accurately predict the thresholds for 16.0 μm fibers (MAE = 5.36×10^{-3} mV, comparable to 5.53×10^{-3} mV on the full dataset), but its performance degraded for 5.7 μm fibers (MAE = 6.77×10^{-2} mV, compared to 1.22×10^{-2} mV when trained on the full dataset). This reduction may be due to the distinct biophysical mechanisms of smaller-diameter axons [58], which are more difficult for the CNN to capture through extrapolation from large-diameter data.

Overall, our findings suggest that CNNs can accurately substitute for computational models in predicting activation thresholds of myelinated axons within a certain diameter range. For smaller-diameter fibers and unmyelinated fibers, once detailed computational models and datasets are established, CNNs or more advanced neural network architectures may be able to capture the underlying membrane dynamics, thereby extending applicability across a wider range of axonal geometries and enabling more generalizable threshold predictions.

D. Input Representations of the CNN

Different from other related works [34], [36], we use the potential distribution directly as the input to the neural network, instead of the second-order spatial derivative, which is usually thought to be directly related to neural activation [28], [59]. On one hand, CNNs are theoretically capable of extracting derivative information from the input [60]. On the other hand, it helps avoid error amplification that can occur during differentiation. Additionally, directly using the potential distribution as input also facilitates normalization for preprocessing.

E. Limitations and Future Work

However, this study has several limitations. The fiber lengths in our dataset range from 20 to 100 mm, and stimulation pulse widths vary from 100 to 1000 μs. While these parameters cover a majority of application scenarios, the prediction accuracy may decline for settings outside these ranges. Moreover, our investigation is confined to monophasic square-wave stimulation, excluding other stimulation modalities such as biphasic stimulation. Nevertheless, the framework established herein provides a basis for constructing corresponding datasets and training relevant neural networks for different stimulation forms. Additionally, we only focused on the response of the MRG model to single-pulse stimulation, lacking the ability to predict activation thresholds under electrical stimulation with multiple pulses at different frequencies.

Future work will aim to broaden the range of applicability by developing datasets that encompass additional stimulation settings and a wider range of nerve fiber geometrical parameters. Besides, more advanced neural network architectures, such as attention-based models, could be explored to further enhance the model's performance. Furthermore, more direct clinical applications should be explored based on the established threshold prediction capability. For instance, this model could assist in parameter-tuning process for diverse neuromodulation scenarios, optimizing the design of neural interfaces, and facilitating more precise and personalized treatment strategies based on the reconstruction of nerve fiber trajectories from clinical imaging (e.g. diffusion tensor imaging technology). Integrating this model into clinical devices to enable rapid visualization of activation states is also an important direction for future work.

V. Conclusion

In this paper, we have established a CNN-based framework that accurately predicts activation thresholds for myelinated axons and demonstrated its applicability across various neuromodulation scenarios. This approach will enable near real-time estimations of neural recruitment under complex stimulation

parameters, demonstrating the potential for clinical application. Our work enables the construction of more precise and complex neural computational models by accelerating the computation process, thus opening a new avenue for the advancement of more personalized and intelligent neuromodulation therapies.

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